

OpenStack Administration & Operations Training

Total Duration: 40 Hours

- Training Duration: 32 Hours
- Final Assessment (MCQ, Lab, Viva): 8 Hours

Prepared using the provided reference with focus on practical implementation, deployment flow, troubleshooting, and enterprise-level understanding.

Day 1 — OpenStack Fundamentals & Architecture

Concepts

- OpenStack ecosystem & OpenInfra Foundation
- OpenStack release lifecycle
- Enterprise OpenStack use cases
- Logical architecture overview
- Control Plane vs Data Plane
- Node types:
 - Controller
 - Compute
 - Network
 - Storage
- Supporting services:
 - RabbitMQ
 - MariaDB
 - Memcached
 - Placement

Lab Exercise

- Map services to architecture
 - Identify services in sample deployment
 - Understand service communication flow
 - Analyze OpenStack architecture diagram
 - Verify service dependencies
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Day 2 — Deployment using Kolla-Ansible

Concepts

- Deployment methods overview
 - DevStack
 - Packstack
 - Kolla-Ansible
 - OpenStack-Ansible
 - Sunbeam
- Production deployment considerations
- Containerized OpenStack deployment architecture

Lab Exercise

Environment Preparation

- Configure controller & compute nodes
- Configure networking prerequisites
- Docker & Python dependency setup

Kolla-Ansible Deployment

- Install Kolla-Ansible
- Configure inventory
- Bootstrap servers
- Perform pre-checks
- Deploy OpenStack cluster

Core Services Configuration

- Keystone configuration
- Glance configuration

- Nova configuration
- Neutron configuration
- Horizon configuration
- Placement configuration
- RabbitMQ configuration
- MariaDB configuration
- Memcached configuration

Validation

- Verify containers
 - Verify OpenStack endpoints
 - Validate OpenStack services
 - Access Horizon dashboard
 - Configure OpenStack CLI
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Day 3 — Keystone & Horizon

Concepts

Keystone

- Authentication vs Authorization
- RBAC in OpenStack
- Service catalog
- Token workflow

Horizon

- Horizon architecture
- Project vs Admin views
- Dashboard workflow

Lab Exercise

Keystone

- Create domains
- Create projects

- Create users
- Assign roles
- Validate tokens
- Verify service catalog
- Test RBAC permissions

Horizon

- Login to Horizon
 - Create keypairs
 - Launch VM from dashboard
 - Verify project resources
 - Monitor instances from Horizon
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Day 4 — Glance & Nova

Concepts

Glance

- Image architecture
- Image formats:
 - QCOW2
 - RAW
 - VMDK
- Image metadata
- Image backends

Nova

- Nova architecture
- Hypervisor basics
- Instance lifecycle
- Flavors & resource allocation
- Availability zones
- Host aggregates
- NUMA & CPU pinning basics

Lab Exercise

Glance

- Upload Linux images
- Manage image visibility
- Inspect image metadata
- Validate image backend

Nova

- Create flavors
 - Launch instances
 - Start/Stop/Reboot VMs
 - Access VM using SSH
 - Verify scheduling
 - Configure availability zones
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Day 5 — Instance Management & Neutron

Concepts

Instance Management

- Instance states
- Snapshots
- Resize workflow
- Migration basics
- Instance logging

Neutron

- Neutron architecture
- Provider vs Tenant networks
- Flat, VLAN & VXLAN networks
- Routers & gateways
- Floating IP concepts
- DHCP & Metadata services

Lab Exercise

Instance Management

- Create snapshots
- Resize instances
- Restore instances
- Analyze instance logs
- Perform instance operations

Neutron

- Create networks & subnets
 - Configure routers
 - Attach instances to networks
 - Configure security groups
 - Assign floating IPs
 - Validate connectivity
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Day 6 — Advanced Networking & Storage

Concepts

Advanced Networking

- L2 & L3 agents
- Network namespaces
- Security group workflow
- SDN overview (concept only)
- Cisco ACI integration concepts (concept only)

Storage Services

Cinder

- Cinder architecture
- Volume lifecycle
- Storage backends

Swift

- Swift architecture

- Rings & replication
- Containers & objects

Lab Exercise

Networking

- Inspect network namespaces
- Validate Neutron agents
- Verify floating IP communication
- Troubleshoot connectivity

Storage

- Create volumes
 - Attach volumes to instances
 - Extend volumes
 - Upload objects to Swift
 - Verify object storage operations
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Day 7 — HA, Monitoring & Security

Concepts

High Availability (Concept only)

- Controller HA architecture
- Galera clustering
- RabbitMQ clustering
- HAProxy & Keepalived

Monitoring & Troubleshooting

- OpenStack log locations
- Service health checks
- Common service failures
- Debugging methodology

Security

- Keystone policies
- TLS & API security
- Tenant isolation
- Barbican overview (concept only)

Lab Exercise

- Inspect Nova logs
 - Inspect Neutron logs
 - Restart OpenStack services
 - Troubleshoot failed VM launch
 - Verify cluster health
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Day 8 — Automation, Migration & Enterprise Design

Concepts

Automation

- OpenStack CLI
- OpenStack SDK
- Ansible automation
- Heat orchestration templates

Migration (Concept only)

- Migration strategies
- VMware to OpenStack migration
- Network & storage mapping
- Hystax migration workflow
- Manual migration understanding
- Understanding and using vm image compatible to openstack
- Tools like qemu-img , v2v etc

Lab Exercise

Automation

- Automate tasks using Ansible
 - Validate Heat template deployment
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Final Assessment — 8 Hours

MCQ Assessment

- OpenStack architecture
- Keystone
- Nova
- Neutron
- Cinder & Swift
- HA & Security
- Automation & Migration

Lab Assessment

- Deploy and validate services
- Launch and manage instances
- Configure networking
- Troubleshoot service failures
- Validate storage operations

Viva / Scenario-Based Discussion

- Enterprise deployment design
- Troubleshooting approach
- Migration planning
- HA architecture discussion
- Real-world operational scenarios